

Other Unlicensed Technologies

WiFi is not alone. Several other unlicensed wireless data technologies are either commercially available or nearly so. Each has technical characteristics that lend themselves to particular market opportunities, though there are many areas of potential overlap. The beauty of an unlicensed environment is that hardware vendors and service providers need not go through a gatekeeper such as a cellular carrier to gain access to spectrum. If the technology works, and there is a market for it, the equipment can simply be deployed.

802.11 Variants

WiFi is a particular protocol designed for local-area network applications. Several companies are trying to marry the cost economies of standards-based 2.4 GHz radios with proprietary software and hardware to support additional capabilities. There are also competing standards to WiFi, including HomeRF and the European HyperLAN2, but these have generally lost out to WiFi in market adoption and will likely fade away.

Ultra-Wideband

As described above, ultra-wideband (UWB) systems use such low power that they can underlay beneath existing licensed spectrum bands. Because of the power limitations, current UWB implementations have limited range, but they offer significant capacity. Vendors such as Time Domain and XTreme Spectrum are building chipsets to deliver 100 Mbps or more over short distances. After a long and bitter fight, the FCC authorized UWB underlay for the first time in February. The FCC's initial rules put strict limits on UWB systems, but the Commission committed to reviewing and potentially loosening the restrictions if interference fears do not materialize.

Bluetooth And Other Personal Area Network Technologies

WiFi works in a local area, but many wireless applications need only a range of a few feet. Personal-area networking (PAN) encompasses situations such as transmitting between a mobile phone